

PYTHON & APIs

TALKING TO THE INTERNET WITH CODE!

Have You Ever Ordered Food Online?

Think about when you open Swiggy or Zomato to order a yummy pizza. You tap a button, and the restaurant instantly knows what you want! But how do they talk to each other? They use something magical called an **API**. Guess what? Python can talk to the internet using APIs too!

1. What is an API?

API stands for *Application Programming Interface*. That sounds super long, but it's just a digital bridge that lets two different computer programs talk to each other!

The Restaurant Waiter Analogy 🍽️

Imagine you are sitting at a restaurant table. You want delicious food from the kitchen, but you aren't allowed to walk inside the kitchen yourself! So, what do you do?

- You tell the **Waiter** what food you want.
- The **Waiter (API)** carries your **Request** to the kitchen server.
- The kitchen cooks the food and hands it to the waiter.
- The **Waiter (API)** brings the food back to you as a **Response**!

In this story: **You** are the Python Program, the **Waiter** is the API, and the **Kitchen** is the Internet Server!

2. Request & Response (The Chai Story) ☕

Every time Python talks to an API, it works in two simple steps:

1. **Request:** You asking the internet for data. (Like shouting: "*Bhaiya, one cutting chai!*")
2. **Response:** The internet giving the data back to you. (Like the tea seller handing you the cup!)

Status Codes: How Did It Go?

Websites tell us how they feel using number codes:

- **Status Code 200:** Success! Your tea is ready! 🎉

- **Status Code 404:** Not Found! Oh no, the tea seller ran out of milk today! 🛑

3. The Requests Library — Your Jio SIM Card 📱

To make phone calls to your friends, your mobile phone needs a **Jio or Airtel SIM card**. Without it, you can't connect! In the exact same way, Python needs a special tool called the **requests library** to connect to the internet.

First, we download it into our computer using this command in the terminal:

```
pip install requests
```

4. Live Coding: Let's Fetch a Random Internet Joke! 🤖

Let's write a real Python program that dials a special internet address (a **URL**) and downloads a funny joke! Follow these three simple steps:

```
# STEP 1: Import the SIM card tool
import requests

# STEP 2: Define the internet phone number (API URL)
url = "https://official-joke-api.appspot.com/random_joke"

# STEP 3: Send the request and grab the response!
response = requests.get(url)
data = response.json() # Converts the web data into a Python dictionary!

# Step 4: Print the funny joke setup and its punchline!
print(data['setup'])
print(data['punchline'])
```

5. What is JSON Data? (The Aadhaar Card Format) 🪪

When the internet waiter brings back data, it arrives formatted in a clean layout called **JSON**. Think of JSON exactly like your **Aadhaar Card**. It holds information neatly using keys and values, looking just like a Python dictionary:

```
{
    'name': 'Priya Sharma',
    'city': 'Bengaluru',
    'role': 'Python Coder'
}
```

6. Vocabulary Flashcards ⚡

API

The digital waiter carrying messages between your code and the internet.

URL / Endpoint

The unique web address or "phone number" where the API lives.

Requests Library

Python's SIM card tool that lets it make web calls.

JSON

Neat text data organized like an Aadhaar card or a Python dictionary.

PRACTICE WORKSHEET

TEST YOUR API SUPERPOWERS! (NO ANSWERS ALLOWED)

Hey Coding Champion! Answer these 12 fun questions to see how well you know internet APIs. Write your answers neatly in the blank lines!

Q1. What do the letters "API" stand for?

Q2. In our restaurant story, which character plays the role of the API? (The Diner, the Waiter, or the Kitchen Chef?)

Q3. When you shout "Bhaiya, one cutting chai!", are you sending an API "Request" or getting an API "Response"?

Q4. Which awesome number code tells you that your API call was a huge success (Status Success)?

Q5. What error code does the internet show you if it cannot find the address you are looking for?

Q6. Which specific Python library acts like a mobile SIM card, allowing your script to connect to online apps?

Q7. Write out the exact word you must type at the very first line of your Python code to bring the requests library into your script.

Q8. What do we call the special internet address link (like a phone number) that tells Python exactly where to go to find an API?

Q9. True or False: JSON data format is messy and hard to read, looking nothing like a Python dictionary.

Q10. Why do we write `response.json()` in our joke code? What does it do to the web response?

Q11. Imagine you run the random joke program 3 separate times. Will it show you the exact same joke text every single time? Explain why.

Q12. If you want to grab data from the internet using an API URL inside Python, which specific function command do you join with requests? (Hint: `requests.____(url)`)
